Practical - 7

PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

A. Theory

1. Is the form submitted using POST?

Answer :

Yes. Both the register.html and contact.html forms were updated to include method="POST" to securely send data to the PHP scripts without exposing data in the URL.

2. Are inputs validated and sanitized server side?

Answer :

Yes. filter_var($email, FILTER_VALIDATE_EMAIL) is used to validate email fields. Text inputs are thoroughly sanitized to prevent Cross-Site Scripting (XSS) by combining trim(), strip_tags(), and htmlspecialchars().

3. Is file writing handled safely?

Answer :

Yes. The script first checks if the DATA directory exists and creates it if it doesn't. When writing to the CSV, it uses fopen($file_path, 'a') to append data safely without overwriting previous records. When writing to JSON, it correctly retrieves the existing data, decodes it, pushes the new record, and encodes it back using file_put_contents.

4. Are success and error responses displayed clearly?

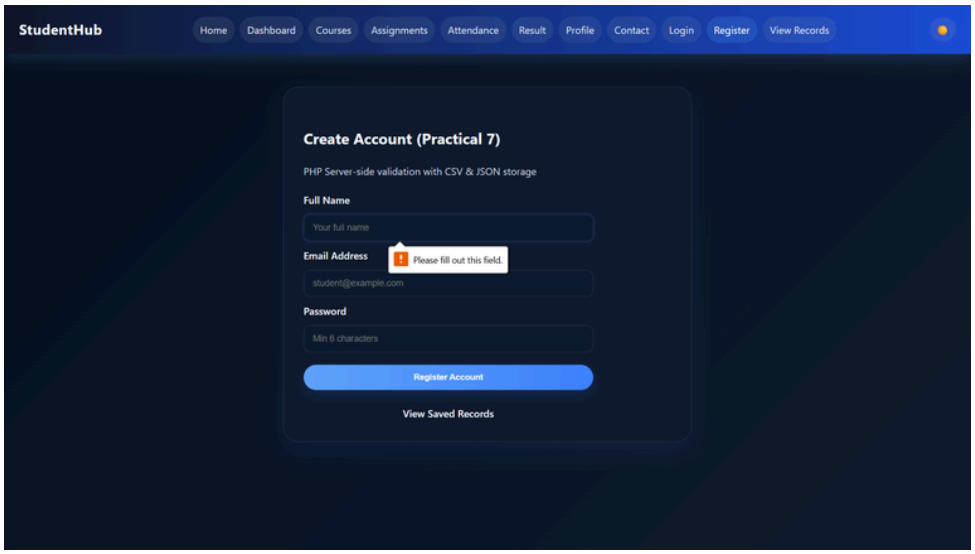
Answer :

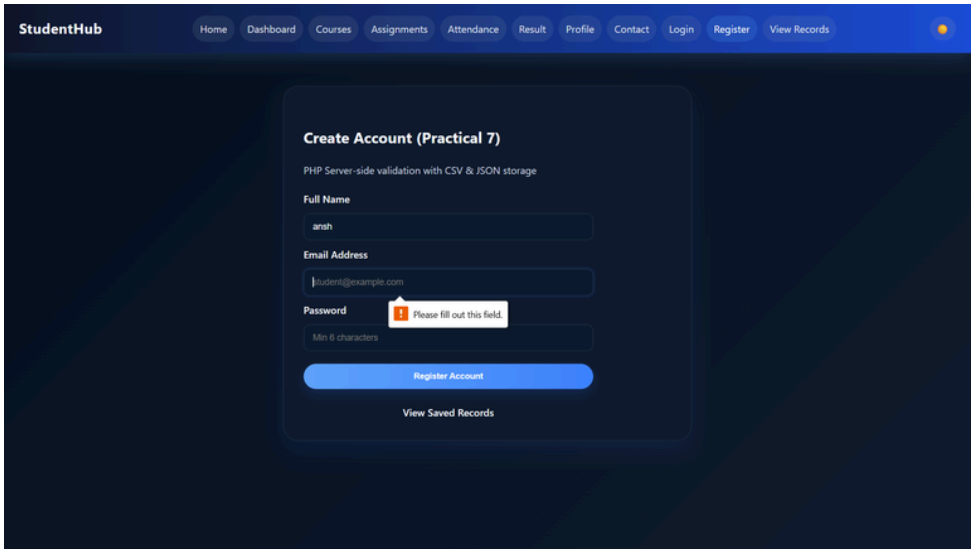
Yes. After processing, the script outputs a styled HTML response. Depending on a $success boolean, the message text color changes dynamically (green #4CAF50 for success, red #f44336 for errors), clearly informing the user of the outcome.

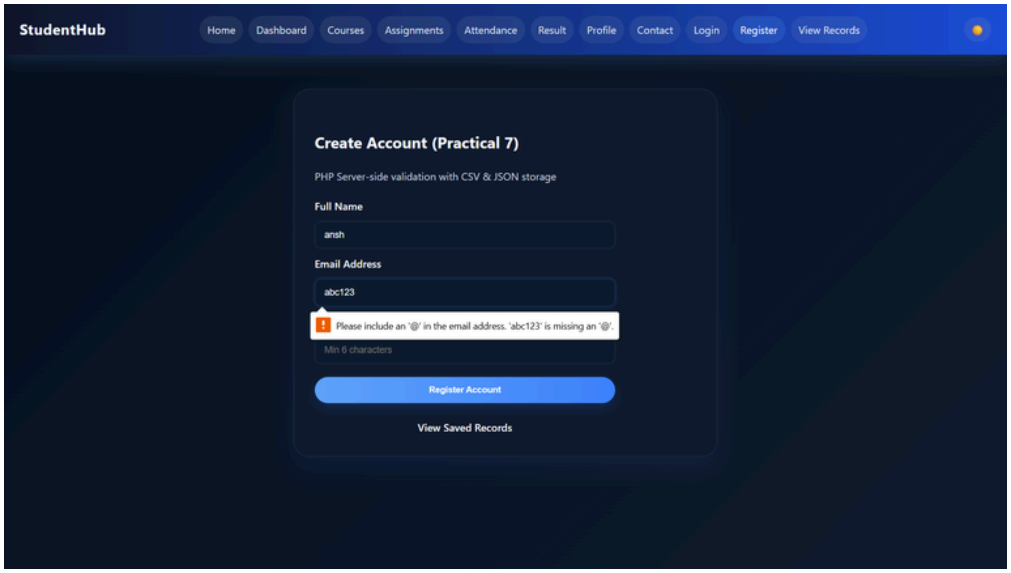
B. Implementation

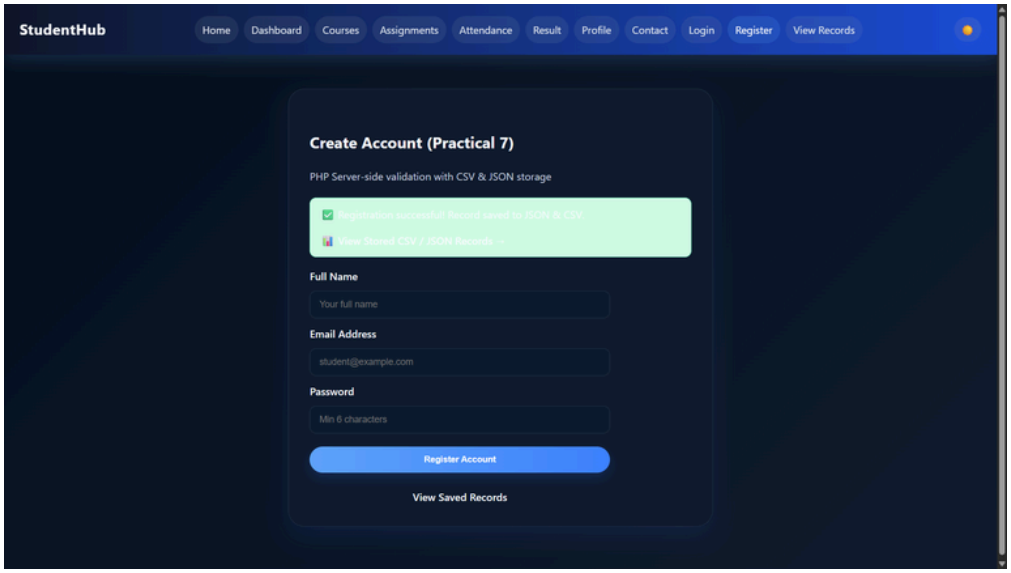
Transitioned the project from purely frontend to full-stack. Configured forms in register.html and contact.html to send POST requests. Created process_register.php (which sanitizes data and saves to a .json file) and process_contact.php (which sanitizes data and appends it to a .csv file).

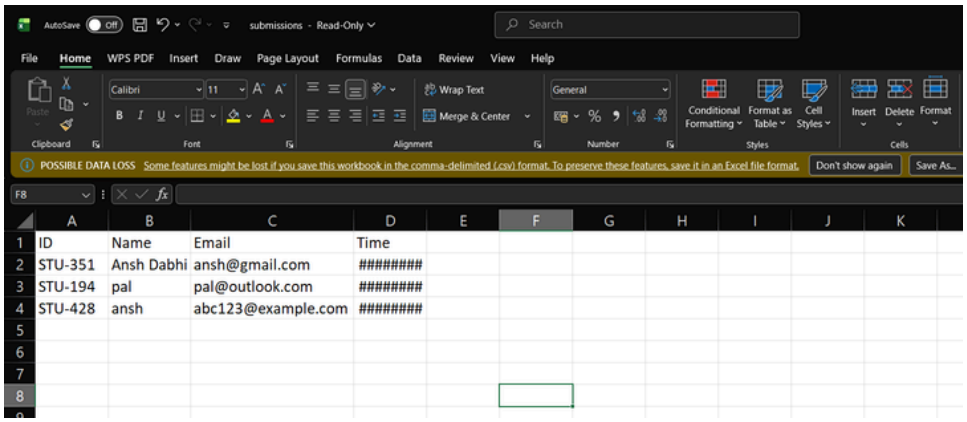
C. Evidence of Work (Screenshots)






StudentHub
Home
Dashboard
Courses
Assignments
Attendance
Result
Profile
Contact
Login
Register
View Records
Create Account (Practical 7)
PHP Server-side validation with CSV & JSON storage
Full Name
ansh
Email Address
abc123
Please include an '@' in the email address. 'abc123' is missing an '@'.
Min 6 characters
Register Account
View Saved Records


StudentHub
Home
Dashboard
Courses
Assignments
Attendance
Result
Profile
Contact
Login
Register
View Records
Create Account (Practical 7)
PHP Server-side validation with CSV & JSON storage
Registration successful! Record saved to JSON & CSV.
View Stored CSV / JSON Records →
Full Name
Your full name
Email Address
student@example.com
Password
Min 6 characters
Register Account
View Saved Records


submissions - Read-Only
Search
File
Home
WPS PDF
Insert
Draw
Page Layout
Formulas
Data
Review
View
Help
Wrap Text
Merge & Center
General
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
POSSIBLE DATA LOSS
Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.
Don't show again
Save As...
F8
ID
Name
Email
Time
STU-351
Ansh Dabhi
ansh@gmail.com
########
STU-194
pal
pal@outlook.com
########
STU-428
ansh
abc123@example.com
########

D. GitHub Repository Link:

https://github.com/AnshDabhi-26/Student-Hub/tree/cbd48695b2c819d31ab4324d32b1d26e698e3512/Practical-7

E. Challenges Faced:

- Understanding the bridge between the frontend (HTML forms) and the backend (PHP $_POST array).
- Dealing with server environments (XAMPP/Apache) and figuring out file permission issues (chmod 777 on macOS) so that PHP had the correct read/write access to create the DATA directory and the JSON/CSV files.
- Ensuring security by heavily sanitizing inputs (htmlspecialchars, strip_tags, FILTER_SANITIZE_EMAIL) to prevent XSS (Cross-Site Scripting) attacks.

F. Learning Outcomes:

Mastered the fundamentals of Backend Web Development: handling HTTP POST requests, server-side data sanitization, file I/O operations in PHP (working with fopen, fputcsv, and json_encode), and running a local server environment (XAMPP).

Faculty Remarks: